Chapter 25: Early Tetrapods and Modern Amphibians

Movement onto Land

- **Physical differences** that animals must accommodate when moving from water to land:
  - **Oxygen content**: Oxygen is at least 20 times more abundant in air and diffuses much more rapidly than in water.
  - **Density**: Air is approximately 1000 times less buoyant and 50 times less viscous than water, requiring strong limbs and skeletal remodeling for structural support.
  - **Temperature regulation**: Air temperature fluctuates more readily than water; terrestrial environments experience harsh cycles of freezing, thawing, drying, and flooding.
  - **Habitat diversity**: Terrestrial environments offer a variety of habitats (forests, grasslands, deserts, etc.) and safe shelter for eggs and young.

Early Evolution of Terrestrial Vertebrates

- By the **Devonian period** (400 million years ago), bony fishes had diversified.
- **Pre-adaptations** for life on land in aquatic ancestors:
  - **Air-filled cavity** (swim bladder/lung): Connected to the pharynx, used for buoyancy or gas exchange.
  - **Paired internal nares** (choanae): Functioned in chemoreception; on land, used to draw air into the air-filled cavity.
  - **Bony elements of paired fins**: Modified for support and movement.
- **Homology**: Lungs and swim bladders are homologous structures.
- **Evolutionary lineage**: Only one transition in the early Devonian provided the ancestral lineage of all tetrapod vertebrates.
- Adaptations included increased vascularization of the air-filled cavity (lung) and **double circulation**.
- **Lobe-finned fishes** are the closest relatives (sister group) of tetrapods.

Figure 1: Tentative cladogram of the Tetrapoda with emphasis on descent of the amphibians. Especially controversial are the relationships of major tetrapod groups (Amniota, Temnospondyll and diverse early tetrapod groups) and outgroups (Actinistia, Dipneusti, extinct Devonian groups). All aspects of this cladogram are controversial, however, including relationships of the Lissamphibia. The relationships shown for the three groups of Lissamphibia are based on molecular evidence. Extinct groups are marked with a dagger symbol (†).

- **Eusthenopteron**: A Devonian lobe-finned fish with an upper arm bone (**humerus**), two forearm bones (**radius** and **ulna**), and wrist-like elements in its fins; could paddle but not walk.
- **Tiktaalik**: Morphologically intermediate between lobe-finned fishes and tetrapods; likely inhabited shallow streams, used limbs to support body to breathe air.
- **Acanthostega**: One of the earliest known Devonian tetrapods; had well-formed tetrapod limbs with 8 digits but limbs were too weak for land travel.
- **Ichthyostega**: Had fully developed shoulder girdle, bulky limb bones, and muscles for terrestrial life; retained tail with fin rays and opercular bones.
- **Pentadactyl plan**: Early tetrapods had more than 5 digits; the 5-digit pattern stabilized later.

Figure 2: Evolution of tetrapod limbs. Limbs of tetrapods evolved from fins of Paleozoic fishes. **Eusthenopteron**, a late Devonian lobe-finned fish (rhipidistian) had paired muscular fins supported by bony elements that foreshadowed the bones of tetrapod limbs. The anterior fin contained an upper arm bone (humerus), two forearm bones (radius and ulna), and smaller elements homologous to wrist bones of tetrapods. As typical of fishes, the pectoral girdle, consisting of the cleithrum, clavicle, and other bones, was firmly attached to the skull. In **Acanthostega**, one of the earliest known Devonian tetrapods (appearing about 360 million years BP), dermal fin rays of the anterior appendage were replaced by eight fully evolved fingers. **Acanthostega** was probably exclusively aquatic because its limbs were too weak for travel on land. **Ichthyostega**, a contemporary of **Acanthostega**, had fully formed tetrapod limbs and must have been able to walk on land. The hindlimb bore seven toes (the number of front limb digits is unknown). **Limnoscelis**, an anthracosaur of the Carboniferous (about 300 million years BP) had five digits on both front and hindlimbs, the basic pentadactyl model that became the tetrapod standard.

- **Temnospondyls**: A group including several extinct lineages and **Lissamphibia** (modern amphibians).
- **Lissamphibia**: Arose during the Carboniferous; ancestors of frogs (**Anura**), salamanders (**Caudata**), and caecilians (**Apoda**).
- **Lepospondyls** and **Anthracosaurs**: Closer to amniotes than to temnospondyls.

Figure 3: Early tetrapod evolution and the descent of amphibians. Tetrapods share most recent common ancestry with diverse Devonian groups. Amphibians share most recent common ancestry with diverse temnospondyls of the Carboniferous and Permian periods of the Paleozoic, and Triassic period of the Mesozoic.

Modern Amphibians

- **General characteristics**:
  - Ectothermic, primitively quadrupedal tetrapods with glandular skin.
  - Many depend on freshwater for reproduction.
  - **Metamorphosis**: Ancestral life history involves aquatic eggs, aquatic larvae with gills, and metamorphosis to terrestrial adults with lungs.
  - **Exceptions**: Some salamanders are permanently aquatic (paedomorphosis); some caecilians and frogs have direct development (no aquatic larval phase).
  - **Skin**: Thin, moist, water-permeable; requires moist environments; susceptible to desiccation.
  - **Ectothermy**: Body temperature determined by environment; restricts range to cool, wet areas.
  - **Excretory system**: Paired **mesonephric** or **opisthonephric** kidneys; **urea** is the main nitrogenous waste.
  - **Cranial nerves**: Ten pairs.

Caecilians: Order Gymnophiona (Apoda)

- **Characteristics**:
  - Elongate, limbless, burrowing.
  - Tropical forests of South America, Africa, India, Southeast Asia.
  - Small dermal scales in some; many vertebrae; long ribs; terminal anus.
  - Small eyes (often blind); sensory tentacles on snout.
- **Reproduction**: Internal fertilization (male has protrusible copulatory organ); eggs deposited in moist ground; some have aquatic larvae, others direct development; viviparity common.

Figure 4: A, Female caecilian coiled around eggs in burrow. B, Pink-head caecilian (**Herpele multiplicata**), native to western Africa.

Salamanders: Order Urodela (Caudata)

- **Characteristics**:
  - Tailed amphibians; Holarctic distribution (abundant in North America).
  - Limbs usually at right angles to trunk; forelimbs and hindlimbs of equal size.
  - Carnivorous as larvae and adults.
  - Ectotherms with low metabolic rate.
- **Life Cycles**:
  - Ancestral: Aquatic larvae and terrestrial adults.
  - **Internal fertilization**: Female recovers **spermatophore** deposited by male.

Figure 5: Courtship and sperm transfer in pygmy salamanders, **Desmognathus wrighti**. After judging the female's receptivity by the presence of her chin on his tail base, the male deposits a spermatophore on the ground, then moves forward a few paces. A, The white mass of the sperm atop a gelatinous base is visible at the level of the female's forelimb. The male moves ahead, the female following until the spermatophore is at the level of her vent. B, The female has recovered the sperm mass in her vent, while the male arches his tail, tilting the female upward and presumably facilitating recovery of the sperm mass. The female later uses the sperm stored in her body to fertilize eggs internally before laying them.

- Aquatic species lay eggs in water (clusters/stringy masses); terrestrial species lay eggs in moist earth (grapelike clusters), often with parental guarding; terrestrial species often have **direct development**.

Figure 6: Female dusky salamander (**Desmognathus** sp.) attending eggs. Some salamanders exercise parental care of eggs, which includes rotating eggs and protecting them from fungal infections and predation by various arthropods and other salamanders.

- **Newts**: Complex life cycle (aquatic larva -> terrestrial “red eft” -> aquatic adult).

Figure 7: Life history of a red-spotted newt, **Notophthalmus viridescens** of the family Salamandridae. In many habitats the aquatic larva metamorphoses into a brightly colored “red eft” stage, which remains on land from 1 to 3 years before transforming into a secondarily aquatic adult.

- **Respiration**:
  - Extensive vascular nets in skin (**cutaneous respiration**).
  - External gills, lungs, both, or neither at various stages.
  - **Amphiumas**: Aquatic but breathe primarily by lungs.
  - **Plethodontidae**: Lungless; strictly terrestrial or aquatic; rely on cutaneous and **buccopharyngeal breathing**.

Figure 8: Longtail salamander, **Eurycea longicauda**, a common plethodontid salamander.

- **Paedomorphosis**: Retention of larval features in adulthood.

- **Perennibranchiate**: Permanently gilled (e.g., **Necturus**).
- **Axolotls** (**Ambystoma**): Gilled individuals that can metamorphose if pond evaporates or if treated with thyroxine (**T4**).
- **Thyroid hormones** (**T3**, **T4**) are essential for metamorphosis.

Figure 9: Paedomorphosis in salamanders. A, The mud puppy **Necturus** sp., is a permanently gilled (perennibranchiate) aquatic form. B, An axolotl (**Ambystoma mexicanum**) may remain permanently gilled, or, should its pond habitat evaporate, metamorphose to a terrestrial form that loses its gills and breathes by lungs. The axolotl shown is an albino form commonly used in laboratory experiments but uncommon in natural populations.

- Paedomorphosis in **Bolitoglossa**: Padlike foot for climbing evolved by truncating digital development.

Figure 10: Foot structure of representatives of three different species of the tropical plethodontid salamander genus **Bolitoglossa**. These specimens have been treated chemically to clear the skin and muscles and to stain the bone red/pink and cartilage blue. The species having the most fully ossified and distinct digits (A, C) live primarily on the forest floor. The species having the padlike foot caused by restricted digital growth (B) climbs smooth leaves and stems using the foot surface to produce suction or adhesion for attachment. The padlike foot evolved by paedomorphosis; it was derived evolutionarily by truncating development of the body, which prevents full digital development.

Frogs and Toads: Order Anura (Salientia)

- **Characteristics**:
  - Absence of tails in adults (**Anura**).
  - Specialized for jumping (**Salientia**).
  - **Larvae (tadpoles)**: Long finned tail, internal and external gills, no legs, herbivorous.
  - **Metamorphosis**: Dramatic transformation to adult.
  - **Families**:
    - **Ranidae**: Common frogs (e.g., Bullfrog, Green frog).
    - **Hylidae**: Tree frogs.

Figure 11: Two common North American frogs. A, Bullfrog, **Rana catesbeiana**, largest American frog and mainstay of the frog-leg epicurean market (family Ranidae). B, Green tree frog, **Hyla cinerea**, a common inhabitant of swamps of the southeastern United States (family Hylidae). Note adhesive pads on the feet.

- **Bufonidae**: True toads; short legs, stout bodies, thick skins with warts.

Figure 12: American toad, **Bufo americanus** (family Bufonidae). This principally nocturnal yet familiar amphibian feeds on large numbers of insect pests and on snails and earthworms. The rough skin contains numerous glands that produce a surprisingly poisonous milky fluid, providing excellent protection from a variety of potential predators.

- **Size**: Range from less than 1 cm (**Eleutherodactylus iberia**) to 30 cm (**Conraua goliath**).

Figure 13: **Conraua (Gigantorana) goliath** (family Ranidae) of West Africa, the world's largest frog. This specimen weighed 3.3 kg (approximately 7½ pounds).

- **Habitats and Distribution**:
  - **Rana**: Temperate and tropical regions; near water.
  - **Declines**: Habitat loss, UV radiation, fungal infections, malformations.
  - **Hibernation**: In soft mud or forest floor; tolerance to freezing (accumulation of glucose and glycerol).
  - **Invasive species**: **Xenopus laevis** (African clawed frog) and **Bufo marinus** (giant toad).

Figure 14: African clawed frog, **Xenopus laevis**. The claws, an unusual feature in frogs, are on the hind feet. This frog has been introduced into California, where it is considered a serious pest.

- **Integument and Coloration**:
  - Skin: Outer stratified **epidermis** (contains **keratin**) and inner spongy **dermis**.

- **Glands**: **Mucous glands** (waterproofing) and **granular serous glands** (poison).
- **Dendrobatidae**: Poison-dart frogs with extremely toxic secretions.

Figure 15: Section through frog skin.

- **Chromatophores**:
  - **Xanthophores**: Yellow, orange, red.
  - **Iridophores**: Silvery, light-reflecting (Tyndall scattering produces blue color).
  - **Melanophores**: Black or brown melanin.

Figure 16: Pigment cells (chromatophores). A, Pigment dispersed. B, Pigment concentrated. The pigment cell does not contract or expand; color effects are produced by streaming of cytoplasm, carrying pigment granules into cell branches for maximum color effect or to the center of the cell for minimum effect. Control over dispersal or concentration of pigment is mostly by light stimuli acting through a pituitary hormone.

- Green color = Yellow pigment + Blue structural color.

- Many frogs can adjust color to match background (**camouflage**).

Figure 17: Cryptic coloration of the gray tree frog, **Hyla versicolor**. Camouflage is so good that presence of this frog usually is disclosed only at night by its resonant, flutelike call.

- **Skeletal and Muscular Systems**:
  - **Endoskeleton**: Bone and cartilage; specialized for jumping and swimming.
  - **Vertebral column**: Rigid frame; 9 trunk vertebrae + **urostyle** (fused caudal vertebrae).
  - **Skull**: Lighter, less ossified, flattened profile.
  - **Limbs**: Pentadactyl foot, four-rayed hand.
  - **Muscles**: Anterior/ventral (protraction/adduction) and posterior/dorsal (retraction/abduction).

Figure 18: Skeleton of a bullfrog, **Rana catesbeiana**.

- **Respiration and Vocalization**:
  - **Respiratory surfaces**: Skin (cutaneous), mouth (buccal), lungs.

- **Positive-pressure breathing**: Air forced into lungs by elevating floor of mouth.
- **Vocal cords**: Located in **larynx**; sound produced by passing air between lungs and **vocal pouches** (resonators).

Figure 19: Breathing in a frog. Frogs are positive-pressure breathers that fill their lungs by forcing air into them. A, Floor of mouth is lowered, drawing air in through nostrils. B, With nostrils closed and glottis open, the frog forces air into its lungs by elevating floor of mouth. C, Mouth cavity rhythmically ventilates for a period. D, Lungs are emptied by contraction of body-wall musculature and by elastic recoil of lungs.

- **Circulation**:
  - **Closed system**: Single pressure pump (heart).
  - **Double circulation**: Separate pulmonary (lungs) and systemic (body) circuits.
  - **Heart**: Two atria, one undivided ventricle.
    - **Sinus venosus**: Receives systemic blood -> Right atrium.
    - **Left atrium**: Receives pulmonary blood.
    - **Spiral valve**: In **conus arteriosus**, helps separate oxygenated and deoxygenated blood.

Figure 20: Structure of a frog heart. Red arrows, oxygenated blood. Blue arrows, deoxygenated blood.

- **Feeding and Digestion**:
  - Adults: Carnivorous; protrusible tongue; teeth on premaxillae, maxillae, vomers (to hold prey).
  - Larvae: Herbivorous; long digestive tract for fermentation.
- **Nervous System and Special Senses**:
  - **Forebrain**: Olfactory center (smell).
  - **Midbrain**: Optic lobes (vision, complex integration).
  - **Hindbrain**: Cerebellum (equilibrium/movement) and medulla (reflexes).

Figure 21: Brain of a frog, dorsal and ventral views.

- **Ear**: **Tympanic membrane** (eardrum), **columella** (stapes), inner ear (**utricle**, **saccule**, **lagena**).

Figure 22: Cutaway of frog head showing ear structure. Sound vibrations are transmitted from the tympanic membrane by way of the columella to the inner ear. The eustachian tube allows pressure equilibration between the tympanic cavity and the pharynx.

- **Vision**: Dominant sense; lachrymal glands and eyelids; cornea is refractive surface; accommodation by moving lens forward; retina with **rods and cones**; **nictitating membrane**.

Figure 23: Amphibian eye.

- **Reproduction**:
  - **Amplexus**: Male clasps female; external fertilization.

Figure 24: A male green tree frog, **Hyla cinerea**, clasps a larger female during breeding season in a South Carolina swamp. Clasping (amplexus) is maintained until the female deposits her eggs. Like most tree frogs, these are capable of rapid and marked color changes; the male here, normally green, has darkened during amplexus.

- **Development**: Zygote -> Cleavage -> Blastula -> Gastrulation -> Embryo -> Tadpole.
- **Tadpole**: Ventral mouth, adhesive disc, spiracle (left side), internal gills.
- **Metamorphosis**: Hindlimbs appear first, tail resorbed, intestine shortens, lungs develop.

Figure 25: Life cycle of a leopard frog.

- **Tropical strategies**: Foam masses, overhanging leaves, damp burrows, bromeliads.
- **Parental care**: Carrying tadpoles (poison-dart frogs), dorsal pouch (marsupial frogs), back brooding (Surinam frogs), vocal pouch (Darwin’s frog).

Figure 26: Unusual reproductive strategies of anurans. A, Female South American pygmy marsupial frog, **Flectonotus pygmaeus**, carries developing larvae in a dorsal pouch. B, Female Surinam frog carries eggs embedded in specialized brooding pouches on the dorsum; froglets emerge and swim away when development is complete. C, Male poison arrow frog, **Phyllobates bicolor**, carries tadpoles adhering to its back. D, Tadpoles of a male Darwin's frog, **Rhinoderma darwinii**, develop into froglets in its vocal pouch. When ready to emerge, a froglet crawls into the parent's mouth, which the parent opens to allow the froglet's escape.